Haisem Naeem

AI Product Engineer | Stockholm, Sweden
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Work authorisation: eligible to work in Sweden (residence permit for studies)

SUMMARY

AI product engineer who ships LLM products end to end: RAG, multi-agent systems, voice AI and full-stack apps. Shipped three AI products from FastAPI backend to Next.js and Flutter front ends and production: a building-code compliance platform, a habit-coaching app and an ergonomic-assessment app. Based in Stockholm, studying for an MSc in AI and Language at Stockholm University; co-author of a survey published in ACM Computing Surveys.

EXPERIENCE

AI Engineer, Metaviz, Lahore, Pakistan — Jul 2025 – Aug 2026

- **Building-code compliance SaaS:** shipped a multi-tenant platform that checks construction drawings across 5 disciplines, saving ~80% of manual compliance-reviewer time per drawing and capping LLM cost at $15 per document. Built the 132-endpoint FastAPI REST API, Celery + Redis job pipeline, AWS Textract OCR, OpenAI GPT-5 checks, Supabase row-level security and a Next.js 15 + React verification UI, with a 47/48 pytest pass rate.
- **Cost optimisation:** designed the platform's hybrid RAG pipeline (ChromaDB vector database, BAAI/bge-base-en-v1.5 768-dim embeddings, BM25 reranking) with confidence-gated LLM orchestration that skips GPT calls on rule-deterministic passes, cutting average inference spend per document by ~40%.
- **AI habit-coaching app:** shipped a Flutter app that verifies users' photo proof with GPT-4o Vision at ~$0.0075/proof and coaches them by real-time voice (OpenAI Realtime API, LiveKit, ElevenLabs, Deepgram) on a FastAPI + Firestore (21 collections) backend; cut per-session token cost by ~50% with prompt caching and capped spend at $150 per user per month.
- **On-device ergonomic-assessment app:** shipped a Flutter 3.9 app that scores posture offline with isolate-based YOLOv8n (TFLite FP16, 6.2 MB) + MediaPipe and GPU / NNAPI / XNNPack delegate fallback at 30–80 ms inference latency; ported the Cornell ROSA algorithm from Python to Dart at ±1-point congruence on a 71-photo golden evaluation set (100% pass).
- **Voice AI agents:** delivered 5 production voice agents (Retell AI, ElevenLabs, GoHighLevel, n8n) for real-estate and SMB clients, replacing ~1 receptionist FTE at one SMB client. Through prompt engineering, cut one client's prompt 80% (16,800 to 3,279 words) with an 11-file semantic knowledge base, raising response-rule compliance from ~25% to 98% and reducing per-call token cost by ~45%.

Python Developer (Internship), WebBuggs, Lahore, Pakistan — Jul 2024 – Jan 2025

- Built a complete Retrieval-Augmented Generation (RAG) pipeline on FastAPI for real-time AI responses against a private document corpus.
- Built a Django backend for live multilingual transcription with subtitle display and an AI meeting assistant.

PROJECTS

Tuberculosis Detection via Cough Sounds (final-year project): Multi-view ensemble (CNNs, XGBoost, Wav2Vec) for TB detection at 98.33% accuracy, with a mobile app for low-resource clinical settings; placed 3rd of 250 teams at the FAST Job Fair 2025.

Smart Summarizer (PyTorch, LoRA, Hugging Face): Fine-tuned LLMs with LoRA for academic paper summarisation; evaluation with ROUGE, BLEU, BERTScore and LLM-as-Judge.

Multi-Agent Research Assistant (LangGraph, OpenAI): LangGraph AI agents for literature search, summarisation and cross-paper comparison.

PUBLICATIONS

Rida Amir, **Haisem Naeem**, Arshad Ali and Maryam Gulzar. "MultiObjective Optimization in Wireless Mesh Networks – A Survey." *ACM Computing Surveys*, vol. 58, no. 11, Article 294, 2026. doi.org/10.1145/3800685

Journal Impact Factor 30.4, ranked 1/146 in Computer Science Theory & Methods (2025, per ACM journal homepage).

EDUCATION

MSc Artificial Intelligence and Language, Stockholm University — Aug 2026 – Jun 2028 (expected)

BS Computer Science, FAST-NUCES, Lahore — Sep 2021 – Jun 2025

CGPA 3.19 / 4.0. Dean's List of Honours three times (Fall 2023, Spring 2024, Spring 2025). Coursework: Deep Learning, Generative AI, Software Engineering, Data Science, AI.

SKILLS

AI / LLM: OpenAI (GPT-4o, GPT-5, Realtime, Vision), Anthropic Claude, LangChain, LangGraph, AG2, RAG, vector databases (ChromaDB, pgvector, Pinecone), BM25, Sentence-Transformers, prompt engineering, LLM evaluation, PyTorch, YOLOv8, MediaPipe, TFLite

Voice AI: Retell AI, ElevenLabs, LiveKit, Deepgram, Twilio, n8n

Languages: Python, TypeScript, JavaScript, Dart, SQL, C++

Full-stack: FastAPI, Django, Flask, Next.js 15, React 19, Node.js, Express, Flutter, SQLAlchemy, Celery

Data, cloud and delivery: PostgreSQL, Supabase, Firestore, Redis, AWS (S3, Textract), Docker, GitHub Actions CI/CD, testing (pytest), Vercel

Human languages: English (fluent, IELTS 7.5), Urdu (native), Swedish (beginner)